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ChemE for Sustainability

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Assessing the feasibility of Corn on the Polylactic Acid Market

Introduction

- The U.S. produces more than **40 million tons of plastic** annually
    - **less than 9% was recycled** [1]
- The US sent **27 million tons of plastic waste to landfills** [1].
- This alarming number calls for the need to properly address plastic waste. It demonstrates that recycling initiatives today are severely impractical and poorly enforced and overall, plastic usage itself should be reduced.

Plastic Waste at the Thilafushi Waste Disposal Site [2]

- These **plastics will not biodegrade**, rather they will take centuries to breakdown into microplastics that will **accumulate in our oceans and environment**.
- Rather than waiting to reach a point of no turning back (though some may argue we have already reached it) we must consider alternatives to the practices and the plastics we use today.
- Is there a **greener, more sustainable option** to the main plastics in use today: Polyethylene Terephthalate (PET), High-Density Polyethylene (HDPE), Low-Density Polyethylene (LDPE), Polyvinyl Chloride (PVC), Polypropylene (PP), Polystyrene (PS)?

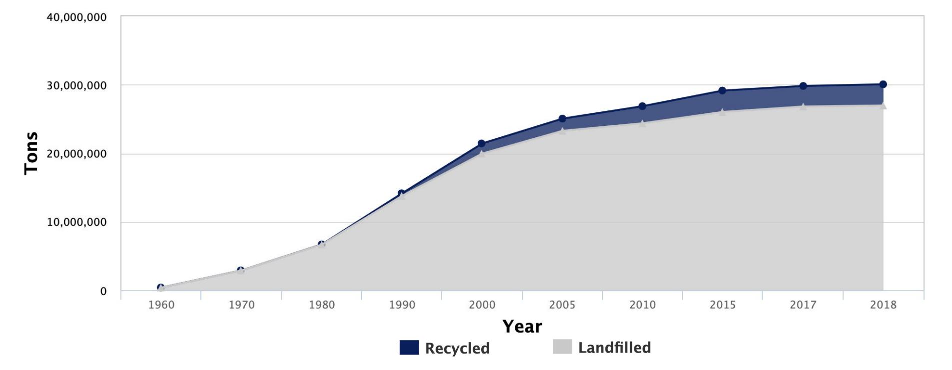
[1] *Plastics: Material-Specific Data*. US EPA. https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data
[2]*Plastic waste continues to rise in 2024*. Sustainable Packaging News. https://spnews.com/plastic-waste-rise/.

Figures on Plastic Production & Waste

Figure 1. Select 2000-2018 Data on Plastic Generation, Recycling, and Landfill in MSW by Weight (in thousands of U.S. tons) [1]

Management Pathway	2000	2005	2010	2015	2017	2018
Generation	25,550	29,380	31,400	34,480	35,410	35,680
Recycled	1,480	1,780	2,500	3,120	3,000	3,090
Landfilled	19,950	23,270	24,370	26,030	26,820	26,970

Figure 2. Plastic Waste Management 1960-2018 [1]



- Increasing trend of plastic generation while plastic **recycling lags severely behind**: under 10% recycling rate
- While the graph & table depict up until 2018, **generation and landfilled plastic projected to increase exponentially** in coming years- these numbers are just the beginning if we keep up current consumption and waste management patterns
- As the rate of generation exceeds the rate of recycling, we are forced into a dangerous situation

[1] *Plastics: Material-Specific Data*. US EPA. https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data.

Which plastics could bio polyesters replace and how large are those markets?

Where PLA (polylactic acid) and PHA (polyhydroxyalkanoates) can compete:

- **PLA is a polyester**, so it has stiffness, clarity, and gas barrier behavior **similar to PET** (polyethylene terephthalate) (Naser et al., 2021).
- **PLA can replace PET in cold drink cups,** clear clamshells**, salad boxes**, and other thermoformed food containers that stay at room or fridge temperatures and do not need very high toughness (Naser et al., 2021).
- **PLA can also replace PS** (polystyrene) and **some PP** (polypropylene) in yogurt cups, coffee capsules, **disposable cutlery**, and nonwoven fibers, where rigidity and shape retention matter more than high heat resistance (Naser et al., 2021).
- **PHA can substitute for PE (polyethylene) or PP in items that need certified compostability**, such as **produce bags**, liners, compostable straws and **cutlery**, and agricultural films (Good Start Packaging, accessed 2025; Naser et al., 2021).

Market sizes today:

- **Polyester fiber** was around **78 million metric tons in 2024** and is about **59 percent of global fiber production** (Textile Exchange, 2025).
- **PET bottles** account for roughly **26.3 million metric tons per year** with a **market value** around **45.6 billion dollars in 2024** (Global Market Insights, 2024).
- **Global plastics production in 2023** is **436 million metric tons** (UNCTAD, 2025).
- **Global bioplastics production is around 1.44 million metric tons**, so **bio polyesters occupy well under one percent of the plastics market** by mass (European Bioplastics, 2024).

PLA share and our stress test scenario

- **PLA currently accounts for only a few tenths of a percent of global plastics by mass** when its output is compared with the 436 million metric ton plastics total (European Bioplastics, 2024; UNCTAD, 2025).

- **Recent literature projects PLA is expected to account for over 20% of the bioplastic market by 2030**, mainly in packaging and fibers that compete with PET, PS, and PP (D'Souza).

- Based on that high growth signal, in this project we assume a stress test where PLA supplies about 20 percent of plastics in its technically suitable applications by 2030, mainly in PET bottles and thermoformed packaging, PS food packaging, and PP packaging and nonwovens.

- **This 20 percent value is our own scenario, built on the market sizes above, and we use it to explore how such growth would affect corn demand and land use** rather than as a literal forecast.

PLA Production Through Corn:

The production of PLA from Corn involves 7 Crucial Steps and Reactions:

Corn $\xrightarrow{1}$ Starch $\xrightarrow{2}$ Glucose $\xrightarrow{3}$ Lactic Acid $\xrightarrow{4}$ Lactide $\xrightarrow{5}$ PLA

Namely, this Process Involves the Following Relevant Chemical Reactions and Conversions:

1. A ~72% loss from converting dry corn to corn starch
2. $(C_6H_{10}O_5)_n + nH_2O \rightarrow n(C_6H_{12}O_6)$
3. $C_6H_{12}O_6 \rightarrow 2CH_3CH(OH)COOH$
4. A ~30 lactic acid to lactide evaporation and reactivity loss lactic
5. $\frac{72.065}{90.081} \approx 0.80 \rightarrow 80\%$

Total conversion efficiency: ~44.9%
*Refer to Appendix 1

[1] "Corn Harvest Quality Report 2019/2020 - Page 6 of 32 - U.S. GRAINS COUNCIL." *U.S. GRAINS COUNCIL*, 22 Apr. 2020, grains.org/corn_report/corn-harvest-quality-report-2019-2020/6/.

[2] Engineering, Chemical. "Technology Profile: Producing Polylactic Acid from Corn." *Chemical Engineering*, 1 May 2021, www.chemengonline.com/technology-profile-producing-polylactic-acid-from-corn/.

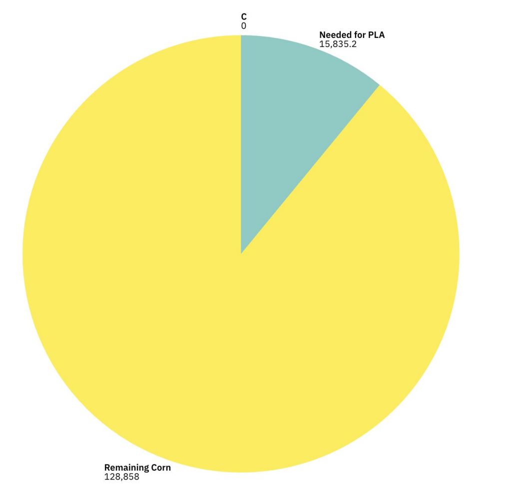
Investments and Demands to Satisfy PLA Creation:

158.352 million metric tons of corn
*Refer to Appendix 2

Land Usage for Corn to Replace PLA Market: 46,697,729 acres

Larger than the entire state of Washington

~12.28% of Corn Market Needed:



[1] De França, Juliene Oliveira Campos, et al. "Polymers Based on PLA from Synthesis Using D,L-Lactic Acid (or Racemic Lactide) and Some Biomedical Applications: A Short Review." *Polymers*, vol. 14, no. 12, 8 June 2022, p. 2317, https://doi.org/10.3390/polym14122317.

[2] Erenstein, Olaf, et al. "Global Maize Production, Consumption and Trade: Trends and R&D Implications." *Food Security*, vol. 14, no. 14, 17 May 2022, link.springer.com/article/10.1007/s12571-022-01288-7, https://doi.org/10.1007/s12571-022-01288-7.

Polyesters

- Polyester is the third most widely used plastic, and it is **produced from crude oil** [1]. Its **tensile strength, durability**, and resistance to shrinking and stretching makes it an essential material across many industries.
- Polyesters such as PET are **economically favorable** as they are relatively inexpensive to manufacture.

- The production of polyester requires non-renewable resources such as fossil fuels, which isn't feasible long term. Polyester is one of the more **carbon-intensive** materials in the textile industry as the production of one kilogram releases, on average, 14.2 kilograms of carbon dioxide [2].
- Though it has the potential to be recycled, polyester will not biodegrade on its own and requires extreme temperatures and pressures. Because PET requires industrial recycling and only 14% of it generated annually is recycled, overall polyester contributes to **significant environmental pollution and plastic waste** [3].

Reliance on fossil fuels is exacerbating the climate crisis [4]

[1]Textile Exchange. *Polyester*. Textile Exchange. https://textileexchange.org/polyester/.
[2]McClements, D. *Polyester: History, Definition, Advantages, and Disadvantages*. www.xometry.com. https://www.xometry.com/resources/materials/polyester/.
[3]*PET Recycling Coalition*. The Recycling Partnership. https://recyclingpartnership.org/pet-recycling-coalition/.
[4]*Synthetics*. Collective Fashion Justice. https://www.collectivefashionjustice.org/synthetics.

Switching to Biopolyesters

- Biopolyesters are not produced from non-renewable resources such as fossil fuels- they are **derived from sustainable resources** such as plants, algae, and microorganisms. This poses a significant advantage as it reduces the reliance on fossil fuels, thus **emitting less greenhouse gasses** in the production process.
- As biopolyesters tend to be less durable, they can be used for single-use items meant to have a shorter lifespan.

Bacteria and algae used for biopolyesters [1]

- While favorable in these aspects, biopolyesters **requires lots of land usage, irrigation systems and water, and fertilizer needs**. To displace polyesters, even marginally, these factors would cause **significant environmental strain**.
- Furthermore, the biopolyesters being less durable pose an issue for items that are multi-use and have longer lifespans. Looking at this from an economic standpoint, the more expensive and underperforming biopolyester is not enough to entice manufactures to make this switch from polyester. Investors would have to expect lots of **risk and price volatility**.
- As with polyester, biopolyester is **not entirely biodegradable** in all environments, such as landfills. Therefore, it still requires industrial recycling infrastructure for complete and effective biodegradation, with a **heavy reliance on people choosing to properly recycle**.

[1] *Scientists Find a Way to Replace Bacteria and Algae in the Production of Bioplastics*. Bioplastics News. https://bioplasticsnews.com/2018/06/22/bioplastics-bacteria-algae-synthetic/ (accessed 2025-12-01).

Potential Next Steps and Areas of Improvement

A major hindrance in switching from polyesters to biopolyesters is the **poor durability and brittleness of the biopolyesters** [1]. To increase the strength and impact resistance, two methods can be used:

- **Nanoparticle incorporation** enables **toughening and compatibilization of PLA** and other brittle biopolyesters through blending ductile polymers or elastomers like natural rubber [2]. Through improving adhesion between phases, materials with desirable properties are created.
- **Plasticizers** are typically used in materials like plastics or concrete to **increase durability and elasticity**. By weakening the bonds between polymer chains, the polymers can move more freely and create more flexible products. Plasticizers include naturally derived compounds like glycerol, epoxidized soybean oil, and castor oil [3].

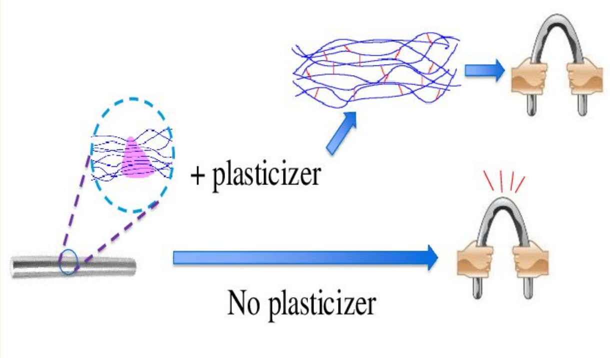


Figure 1. Illustration of the role of a plasticizer in a polymer matrix [4]

These methods provide the **basis of improvement for the quality and desirability** of biopolyesters. Through this, its market can increase and slowly work to displace polyesters with a sustainable and greener alternative that performs well.

[1]Giulia Fredi; Dorigato, A. Compatibilization of Biopolymer Blends: A Review. *Advanced Industrial and Engineering Polymer Research* 2023, *7* (4). https://doi.org/10.1016/j.aiepr.2023.11.002.
[2] Andrade, A.; Oliveira, E. Effect of Nanoparticles as Reinforcements and Compatibilizers of Poly(Lactide)/Poly(Butylene Adipate-Co-Terephthalate) Blends for Application in Food Packaging. *Polymers for Advanced Technologies* 2025, *36* (8). https://doi.org/10.1002/pat.70298.
[3] *What Are Plasticizers? | Plasticizers in Plastic Manufacturing*. Advanced Plastiform, Inc. https://advancedplastiform.com/what-are-plasticizers/.
[4] Saabome, Samuel & Abdul-Malik, Salim Umar & Adu-Poku, Brolin & Yoon, Soongseok. (2020). Title: A Review on Plasticizers and Eco-Friendly Bioplasticizers: Biomass Sources and Market. International Journal of Engineering Research and. V9. 10.17577/IJERTV9IS050788.

D'Souza, Alan, and Rajeev Singh. "Global Polylactic Acid Market Size, Share & Industry Trends Analysis Report by End-Use (Packaging, Agriculture, Electronics, Textile, Automotive & Transport, and Others), by Regional Outlook and Forecast, 2023 - 2030." *KBV Research*, 31 Oct. 2023, www.kbvresearch.com/polylactic-acid-market/.

Conclusion

The PLA market is one of the **fastest growing and promising fields** of plastic-manufacturing. With a 2030 projection to tople $4 billion, PLA is expected to account for over **20% of the bioplastic market** *(D'Souza)*. As such, developing a concrete and sustainable pipeline to produce PLA reliably is crucial in order to continue our global sustainability goals.

Biopolyesters and biodegradable polymers such as PLA offer a potential greener alternative to the current widely used plastics such as PET and PP. While promising, to displace the PET demand and replace with PLA with current methods, the amount of corn needed would be inefficient and **cause enormous environmental strain** due to the land, water, and fertilizer required.

Metric:	Direct Corn to PLA Efficiency	Total Corn to Replace Entire PLA Market Demand:	Total Land Mass Needed Given Modern Corn Growth Spacings
Units:	%	Million metric tons	Acres
Estimation:	44.9	158.352	46,697,729

Summary of Estimation:

While producing and marketing PLA at this scale may not be feasible today, **advancements in PLA yield** and bioplastics through nanoparticle and plasticizer incorporation offer **potential solutions**.

Appendix 1: Corn to PLA Efficiency Conversion Representation:

Take a 1,000 kg Sample of Corn:

$$1,000kg\ Dry\ Corn \times 0.723 = 723kg\ Starch$$

$$\frac{Starch\ Molar\ Mass}{Glucose\ Molar\ Mas} = \frac{180.16}{162.1} \approx 1.11 \Rightarrow\Rightarrow 723 \times 1.11 = 802.53kg\ Glucose$$

$$802.53 \times 1 = 802.53kg\ Lactic\ Acid$$

$$\frac{Lactic\ Acid\ Molar\ Mass}{PLA\ (Repeating)\ Molar\ Mass} = \frac{90.081}{72.065} \approx 0.80 \Rightarrow\Rightarrow 802.53 \times 0.80 = \underline{642.024\text{kg PLA}}$$

Thus, a 1,000 kg sample of Corn will provide a <u>*theoretical*</u> yield of ~642 kg Polylactic Acid.

Accounting for a practical efficiency loss of 30% for unwanted byproducts, heat loss, and fermentation/purification un-optimizations results in a final yield % of **44.9%**

$$642.024 \times 0.7 = 449.4168kg\ of\ PLA \Longrightarrow \underline{\sim 0.4494}$$

Appendix 2: Global Demands Given Estimations:

Annual Polyester Production ≈ ***71.1 million metric tons***

Using our estimated conversion percentage of 44.9%, we find the necessary quantity of corn to completely replace the PLA industry to be:

$$71.1 \times \frac{1}{0.449} =$$ **158.352 million metric tons of corn**

Annual Global Corn Production≈ **1.288 billion metric tons**

$$\frac{158.352}{1{,}288.58} =$$ 0.1228887613. Or just over **12%** of the global corn market.

In terms of expected land usage:

Global land dedication to corn: ≈**380 million acres**

USA land dedication to corn: ≈**96 million acres**

$$380{,}000{,}000 \times 0.1289 =$$ **46,697,729 acres** of land across the world

$=$ **12,374,400 acres** of land in the US

[1] De França, Juliene Oliveira Campos, et al. "Polymers Based on PLA from Synthesis Using D,L-Lactic Acid (or Racemic Lactide) and Some Biomedical Applications: A Short Review." *Polymers*, vol. 14, no. 12, 8 June 2022, p. 2317, https://doi.org/10.3390/polym14122317.

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[3] Ates, Aaron. "USDA ERS - Feed Grains Sector at a Glance." *Www.ers.usda.gov*, USDA, 5 Jan. 2025, www.ers.usda.gov/topics/crops/corn-and-other-feed-grains/feed-grains-sector-at-a-glance.

Thank You!